

MICHAEL ODHIAMBO

Software Engineer, Backend, Mobile & AI Systems

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SUMMARY

Computer Science graduate and full-stack software engineer with a track record of shipping production backend systems, Android apps, and AI-integrated products. Two-time fintech/AI hackathon winner. Comfortable owning a project end to end - architecture, implementation, deployment, and infrastructure - across Kotlin, Python, and JavaScript/TypeScript ecosystems.

TECHNICAL SKILLS

- **Languages:** Kotlin, Python, TypeScript/JavaScript, SQL
- **Backend & APIs:** Ktor, FastAPI, Django, Flask, REST API design, JWT/RBAC auth, PostgreSQL, Redis, webhooks
- **Mobile & Frontend:** Native Android (Kotlin, Jetpack Compose, MVVM, Clean Architecture), React.js, Vite, Tailwind CSS
- **AI & Systems:** LLM integration & function calling, autonomous agent design, Constraint Satisfaction Problem (CSP) algorithms
- **Infrastructure & DevOps:** Docker, nginx/Traefik reverse proxies, VPS deployment, CI pipelines, domain & relay server configuration

KEY PROJECTS

Gatekeeperd - Payment Gating Engine for Self-Hosted Client Apps 2026

Kotlin (Ktor), PostgreSQL, Redis, Docker, nginx/Traefik, Paystack API

- Built a backend service that gates live traffic to self-hosted client applications behind a payment status check, integrated with nginx auth_request and Traefik ForwardAuth.
- Designed a Redis-cached, Postgres-backed gate check (60s TTL) with a configurable fail-open/fail-closed mode for infrastructure outages.
- Integrated Paystack for checkout and webhook-driven payment confirmation, with automatic re-blocking on payment reversal.
- Built an admin API (JWT-secured) for project CRUD, Docker container lifecycle management, and nginx site configuration, including wizard-style multi-step flows with validation-before-apply.
- Implemented a scheduled auto-blocker job that suspends overdue projects after a configurable grace period.

Vela - Remote Laptop Control Platform 2026 – Present

Python (Flask), React.js, Kotlin (Android), VPSS relay, LLM function calling

- Built a cross-platform remote device control system with a React web client and a native Kotlin Android app, connected through a self-hosted relay server.
- Designed and implemented the Flask backend handling secure relay communication between paired devices.
- Integrated an LLM with function calling so natural-language commands trigger real device actions without a hand-written parser.
- Owned the full deployment pipeline: hosting, relay configuration, and domain assignment.

AI Credit Scoring Agent - Ubunifu Hackathon, 1st Place December 2025

LLM integration, fintech data processing, mobile

- Built a credit scoring application that uses informal financial signals - airtime purchases, informal loans, micro-transactions - to generate AI-powered creditworthiness scores for unbanked and underbanked users.

- Owned the AI integration and data pipeline components of the solution.

M-Pesa Expenditure Analyzer - Build With AI Hackathon, 1st Place *December 2025*

AI agent, M-Pesa statement parsing, SMS analysis

- Won the Build With AI hackathon (Unstacked Labs certificate) with a working AI agent that parses M-Pesa statements and SMS transaction messages to assess spending patterns and creditworthiness.
- Rapidly prototyped and shipped a working agent within the hackathon timeline.
- Applied structured data extraction to unstructured mobile financial data to produce actionable financial insights.

Tessera - Automated Timetable Scheduler *2025 – Present*

Constraint Satisfaction Problem (CSP), algorithm design

- Engineered a scheduling system using CSP algorithms to automatically assign rooms, instructors, and time slots while satisfying all academic constraints.
- Currently piloted and tested in production by a school institution.

Raylux Solar - Client Website *2026*

React.js, HTML/CSS, deployment, domain management

- Designed, built, and deployed a production website for a solar installation services client, including domain configuration.
- Managed the engagement end to end - requirements, design, development, and delivery on schedule.

CERTIFICATIONS & COMPETITIONS

- Unstacked Labs Certificate - Build With AI Hackathon, 1st Place (November 2025)
- Certificate of Participation - Zindi Hackathon: fine-tuning ML models for real-world NLP/AI tasks
- Google Developer Profile - me.developers.google.com/u/mikesplore
- Active DEV Community author - technical articles on Kotlin, APIs, and software architecture

EDUCATION

Technical University of Mombasa - Mombasa, Kenya

Bachelor of Science in Computer Science (Awaiting Graduation)

ADDITIONAL INFORMATION

- **Work style:** Remote-friendly, disciplined, detail-oriented
- **Languages:** English (fluent), Swahili (native)
- **Location:** Mombasa, Kenya - open to relocation or remote work